14.1 Basic configuration

The UART0/2/3 peripherals are configured using the following registers:

1. Power: In the PCONP register (Table 46), set bits PCUART0/2/3.

   Remark: On reset, UART0 is enabled (PCUART0 = 1), and UART2/3 are disabled (PCUART2/3 = 0).
2. Peripheral clock: In the PCLKSEL0 register (Table 40), select PCLK_UART0; in the PCLKSEL1 register (Table 41), select PCLK_UART2/3.
3. Baud rate: In register U0/2/3LCR (Table 280), set bit DLAB =1. This enables access to registers DLL (Table 274) and DLM (Table 275) for setting the baud rate. Also, if needed, set the fractional baud rate in the fractional divider register (Table 286).
4. UART FIFO: Use bit FIFO enable (bit 0) in register U0/2/3FCR (Table 279) to enable FIFO.
5. Pins: Select UART pins through the PINSEL registers and pin modes through the PINMODE registers (Section 8.5).

   Remark: UART receive pins should not have pull-down resistors enabled.
6. Interrupts: To enable UART interrupts set bit DLAB =0 in register U0/2/3LCR (Table 280). This enables access to U0/2/3IER (Table 276). Interrupts are enabled in the NVIC using the appropriate Interrupt Set Enable register.
7. DMA: UART0/2/3 transmit and receive functions can operate with the GPDMA controller (see Table 544).

14.2 Features

- Data sizes of 5, 6, 7, and 8 bits.
- Parity generation and checking: odd, even mark, space or none.
- One or two stop bits.
- 16 byte Receive and Transmit FIFOs.
- Built-in baud rate generator, including a fractional rate divider for great versatility.
- Supports DMA for both transmit and receive.
- Auto-baud capability
- Break generation and detection.
- Multiprocessor addressing mode.
- IrDA mode to support infrared communication.
- Support for software flow control.

14.3 Pin description

Table 270: UARTn Pin description

Pin	Type	Description
RXD0, RXD2, RXD3	Input	**Serial Input.** Serial receive data.
TXD0, TXD2, TXD3	Output	**Serial Output.** Serial transmit data.

14.4 Register description

Each UART contains registers as shown in Table 271. The Divisor Latch Access Bit (DLAB) is contained in UnLCR7 and enables access to the Divisor Latches.

Table 271. UART0/2/3 Register Map

Generic Name	Description	Access	Reset value[1]	UARTn Register Name & Address
RBR (DLAB =0)	Receiver Buffer Register. Contains the next received character to be read.	RO	NA	U0RBR - 0x4000 C000 U2RBR - 0x4009 8000 U3RBR - 0x4009 C000
THR (DLAB =0)	Transmit Holding Register. The next character to be transmitted is written here.	WO	NA	U0THR - 0x4000 C000 U2THR - 0x4009 8000 U3THR - 0x4009 C000
DLL (DLAB =1)	Divisor Latch LSB. Least significant byte of the baud rate divisor value. The full divisor is used to generate a baud rate from the fractional rate divider.	R/W	0x01	U0DLL - 0x4000 C000 U2DLL - 0x4009 8000 U3DLL - 0x4009 C000
DLM (DLAB =1)	Divisor Latch MSB. Most significant byte of the baud rate divisor value. The full divisor is used to generate a baud rate from the fractional rate divider.	R/W	0x00	U0DLM - 0x4000 C004 U2DLM - 0x4009 8004 U3DLM - 0x4009 C004
IER (DLAB =0)	Interrupt Enable Register. Contains individual interrupt enable bits for the 7 potential UART interrupts.	R/W	0x00	U0IER - 0x4000 C004 U2IER - 0x4009 8004 U3IER - 0x4009 C004
IIR	Interrupt ID Register. Identifies which interrupt(s) are pending.	RO	0x01	U0IIR - 0x4000 C008 U2IIR - 0x4009 8008 U3IIR - 0x4009 C008
FCR	FIFO Control Register. Controls UART FIFO usage and modes.	WO	0x00	U0FCR - 0x4000 C008 U2FCR - 0x4009 8008 U3FCR - 0x4009 C008
LCR	Line Control Register. Contains controls for frame formatting and break generation.	R/W	0x00	U0LCR - 0x4000 C00C U2LCR - 0x4009 800C U3LCR - 0x4009 C00C
LSR	Line Status Register. Contains flags for transmit and receive status, including line errors.	RO	0x60	U0LSR - 0x4000 C014 U2LSR - 0x4009 8014 U3LSR - 0x4009 C014
SCR	Scratch Pad Register. 8-bit temporary storage for software.	R/W	0x00	U0SCR - 0x4000 C01C U2SCR - 0x4009 801C U3SCR - 0x4009 C01C
ACR	Auto-baud Control Register. Contains controls for the auto-baud feature.	R/W	0x00	U0ACR - 0x4000 C020 U2ACR - 0x4009 8020 U3ACR - 0x4009 C020
ICR	IrDA Control Register. Enables and configures the IrDA mode.	R/W	0x00	U0ICR - 0x4000 C024 U2ICR - 0x4009 8024 U3ICR - 0x4009 C024
FDR	Fractional Divider Register. Generates a clock input for the baud rate divider.	R/W	0x10	U0FDR - 0x4000 C028 U2FDR - 0x4009 8028 U3FDR - 0x4009 C028
TER	Transmit Enable Register. Turns off UART transmitter for use with software flow control.	R/W	0x80	U0TER - 0x4000 C030 U2TER - 0x4009 8030 U3TER - 0x4009 C030

[1] Reset Value reflects the data stored in used bits only. It does not include reserved bits content.

14.4.1 UARTn Receiver Buffer Register (U0RBR - 0x4000 C000, U2RBR - 0x4009 8000, U3RBR - 0x4009 C000 when DLAB = 0)

The UnRBR is the top byte of the UARTn Rx FIFO. The top byte of the Rx FIFO contains the oldest character received and can be read via the bus interface. The LSB (bit 0) represents the “oldest” received data bit. If the character received is less than 8 bits, the unused MSBs are padded with zeroes.

The Divisor Latch Access Bit (DLAB) in LCR must be zero in order to access the UnRBR. The UnRBR is always read-only.

Since PE, FE and BI bits correspond to the byte sitting on the top of the RBR FIFO (i.e. the one that will be read in the next read from the RBR), the right approach for fetching the valid pair of received byte and its status bits is first to read the content of the U0LSR register, and then to read a byte from the UnRBR.

Table 272: UARTn Receiver Buffer Register (U0RBR - address 0x4000 C000, U2RBR - 0x4009 8000, U3RBR - 04009 C000 when DLAB = 0) bit description

Bit	Symbol	Description	Reset Value
7:0	RBR	The UARTn Receiver Buffer Register contains the oldest received byte in the UARTn Rx FIFO.	Undefined
31:8	-	Reserved, the value read from a reserved bit is not defined.	NA

14.4.2 UARTn Transmit Holding Register (U0THR - 0x4000 C000, U2THR - 0x4009 8000, U3THR - 0x4009 C000 when DLAB = 0)

The UnTHR is the top byte of the UARTn TX FIFO. The top byte is the newest character in the TX FIFO and can be written via the bus interface. The LSB represents the first bit to transmit.

The Divisor Latch Access Bit (DLAB) in UnLCR must be zero in order to access the UnTHR. The UnTHR is always write-only.

Table 273: UARTn Transmit Holding Register (U0THR - address 0x4000 C000, U2THR - 0x4009 8000, U3THR - 0x4009 C000 when DLAB = 0) bit description

Bit	Symbol	Description	Reset Value
7:0	THR	Writing to the UARTn Transmit Holding Register causes the data to be stored in the UARTn transmit FIFO. The byte will be sent when it reaches the bottom of the FIFO and the transmitter is available.	NA
31:8	-	Reserved, user software should not write ones to reserved bits.	NA

14.4.3 UARTn Divisor Latch LSB register (U0DLL - 0x4000 C000, U2DLL - 0x4009 8000, U3DLL - 0x4009 C000 when DLAB = 1) and UARTn Divisor Latch MSB register (U0DLM - 0x4000 C004, U2DLL - 0x4009 8004, U3DLL - 0x4009 C004 when DLAB = 1)

The UARTn Divisor Latch is part of the UARTn Baud Rate Generator and holds the value used, along with the Fractional Divider, to divide the APB clock (PCLK) in order to produce the baud rate clock, which must be 16× the desired baud rate. The UnDLL and UnDLM registers together form a 16-bit divisor where UnDLL contains the lower 8 bits of the divisor and UnDLM contains the higher 8 bits of the divisor. A 0x0000 value is treated like a 0x0001 value as division by zero is not allowed. The Divisor Latch Access Bit (DLAB) in UnLCR must be one in order to access the UARTn Divisor Latches. Details on how to select the right value for UnDLL and UnDLM can be found later in this chapter, see Section 14.4.12.

Table 274: UARTn Divisor Latch LSB register (U0DLL - address 0x4000 C000, U2DLL - 0x4009 8000, U3DLL - 0x4009 C000 when DLAB = 1) bit description

Bit	Symbol	Description	Reset Value
7:0	DLLSB	The UARTn Divisor Latch LSB Register, along with the UnDLM register, determines the baud rate of the UARTn.	0x01
31:8	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

Table 275: UARTn Divisor Latch MSB register (U0DLM - address 0x4000 C004, U2DLM - 0x4009 8004, U3DLM - 0x4009 C004 when DLAB = 1) bit description

Bit	Symbol	Description	Reset Value
7:0	DLMSB	The UARTn Divisor Latch MSB Register, along with the U0DLL register, determines the baud rate of the UARTn.	0x00
31:8	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

14.4.4 UARTn Interrupt Enable Register (U0IER - 0x4000 C004, U2IER - 0x4009 8004, U3IER - 0x4009 C004 when DLAB = 0)

The UnIER is used to enable the three UARTn interrupt sources.

Table 276: UARTn Interrupt Enable Register (U0IER - address 0x4000 C004, U2IER - 0x4009 8004, U3IER - 0x4009 C004 when DLAB = 0) bit description

Bit	Symbol	Value	Description	Reset Value
0	RBR Interrupt Enable		Enables the Receive Data Available interrupt for UARTn. It also controls the Character Receive Time-out interrupt.	0
		0	Disable the RDA interrupts.	
		1	Enable the RDA interrupts.	
1	THRE Interrupt Enable		Enables the THRE interrupt for UARTn. The status of this can be read from UnLSR[5].	0
		0	Disable the THRE interrupts.	
		1	Enable the THRE interrupts.	
2	RX Line Status Interrupt Enable		Enables the UARTn RX line status interrupts. The status of this interrupt can be read from UnLSR[4:1].	0
		0	Disable the RX line status interrupts.	
		1	Enable the RX line status interrupts.	
7:3	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
8	ABEOIntEn		Enables the end of auto-baud interrupt.	0
		0	Disable end of auto-baud Interrupt.	
		1	Enable end of auto-baud Interrupt.	

Table 276: UARTn Interrupt Enable Register (U0IER - address 0x4000 C004, U2IER - 0x4009 8004, U3IER - 0x4009 C004 when DLAB = 0) bit description

Bit	Symbol	Value	Description	Reset Value
9	ABTOIntEn		Enables the auto-baud time-out interrupt.	0
		0	Disable auto-baud time-out Interrupt.	
		1	Enable auto-baud time-out Interrupt.	
31:10	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

14.4.5 UARTn Interrupt Identification Register (U0IIR - 0x4000 C008, U2IIR - 0x4009 8008, U3IIR - 0x4009 C008)

The UnIIR provides a status code that denotes the priority and source of a pending interrupt. The interrupts are frozen during an UnIIR access. If an interrupt occurs during an UnIIR access, the interrupt is recorded for the next UnIIR access.

Table 277: UARTn Interrupt Identification Register (U0IIR - address 0x4000 C008, U2IIR - 0x4009 8008, U3IIR - 0x4009 C008) bit description

Bit	Symbol	Value	Description	Reset Value
0	IntStatus		Interrupt status. Note that UnIIR[0] is active low. The pending interrupt can be determined by evaluating UnIIR[3:1].	1
		0	At least one interrupt is pending.	
		1	No interrupt is pending.	
3:1	IntId		Interrupt identification. UnIER[3:1] identifies an interrupt corresponding to the UARTn Rx or TX FIFO. All other combinations of UnIER[3:1] not listed below are reserved (000,100,101,111).	0
		011	1 - Receive Line Status (RLS).	
		010	2a - Receive Data Available (RDA).	
		110	2b - Character Time-out Indicator (CTI).	
		001	3 - THRE Interrupt	
5:4	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
7:6	FIFO Enable		Copies of UnFCR[0].	0
8	ABEOInt		End of auto-baud interrupt. True if auto-baud has finished successfully and interrupt is enabled.	0
9	ABTOInt		Auto-baud time-out interrupt. True if auto-baud has timed out and interrupt is enabled.	0
31:10	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

Bit UnIIR[9:8] are set by the auto-baud function and signal a time-out or end of auto-baud condition. The auto-baud interrupt conditions are cleared by setting the corresponding Clear bits in the Auto-baud Control Register.

If the IntStatus bit is 1 no interrupt is pending and the IntId bits will be zero. If the IntStatus is 0, a non auto-baud interrupt is pending in which case the IntId bits identify the type of interrupt and handling as described in Table 278. Given the status of UnIIR[3:0], an interrupt handler routine can determine the cause of the interrupt and how to clear the active interrupt. The UnIIR must be read in order to clear the interrupt prior to exiting the Interrupt Service Routine.

The UARTn RLS interrupt (UnIIR[3:1] = 011) is the highest priority interrupt and is set whenever any one of four error conditions occur on the UARTn Rx input: overrun error (OE), parity error (PE), framing error (FE) and break interrupt (BI). The UARTn Rx error condition that set the interrupt can be observed via UnLSR[4:1]. The interrupt is cleared upon an UnLSR read.

The UARTn RDA interrupt (UnIIR[3:1] = 010) shares the second level priority with the CTI interrupt (UnIIR[3:1] = 110). The RDA is activated when the UARTn Rx FIFO reaches the trigger level defined in UnFCR[7:6] and is reset when the UARTn Rx FIFO depth falls below the trigger level. When the RDA interrupt goes active, the CPU can read a block of data defined by the trigger level.

The CTI interrupt (UnIIR[3:1] = 110) is a second level interrupt and is set when the UARTn Rx FIFO contains at least one character and no UARTn Rx FIFO activity has occurred in 3.5 to 4.5 character times. Any UARTn Rx FIFO activity (read or write of UARTn RSR) will clear the interrupt. This interrupt is intended to flush the UARTn RBR after a message has been received that is not a multiple of the trigger level size. For example, if a peripheral wished to send a 105 character message and the trigger level was 10 characters, the CPU would receive 10 RDA interrupts resulting in the transfer of 100 characters and 1 to 5 CTI interrupts (depending on the service routine) resulting in the transfer of the remaining 5 characters.

Table 278: UARTn Interrupt Handling

U0IIR[3:0] value[1]	Priority	Interrupt Type	Interrupt Source	Interrupt Reset
0001	-	None	None	-
0110	Highest	RX Line Status / Error	OE[2] or PE[2] or FE[2] or BI[2]	UnLSR Read[2]
0100	Second	RX Data Available	Rx data available or trigger level reached in FIFO (UnFCR0=1)	UnRBR Read[3] or UARTn FIFO drops below trigger level
1100	Second	Character Time-out indication	Minimum of one character in the Rx FIFO and no character input or removed during a time period depending on how many characters are in FIFO and what the trigger level is set at (3.5 to 4.5 character times). The exact time will be: [(word length) $\times$ 7 - 2] $\times$ 8 + [(trigger level - number of characters) $\times$ 8 + 1] RCLKs	UnRBR Read[3]
0010	Third	THRE	THRE[2]	UnIIR Read (if source of interrupt) or THR write[4]

[1] Values "0000", “0011”, “0101”, “0111”, “1000”, “1001”, “1010”, “1011”,”1101”,”1110”,”1111” are reserved.

[2] For details see Section 14.4.8 “UARTn Line Status Register (U0LSR - 0x4000 C014, U2LSR - 0x4009 8014, U3LSR - 0x4009 C014)”

[3] For details see Section 14.4.1 “UARTn Receiver Buffer Register (U0RBR - 0x4000 C000, U2RBR - 0x4009 8000, U3RBR - 0x4009 C000 when DLAB = 0)”

[4] For details see Section 14.4.5 “UARTn Interrupt Identification Register (U0IIR - 0x4000 C008, U2IIR - 0x4009 8008, U3IIR - 0x4009 C008)” and Section 14.4.2 “UARTn Transmit Holding Register (U0THR - 0x4000 C000, U2THR - 0x4009 8000, U3THR - 0x4009 C000 when DLAB = 0)”

The UARTn THRE interrupt (UnIIR[3:1] = 001) is a third level interrupt and is activated when the UARTn THR FIFO is empty provided certain initialization conditions have been met. These initialization conditions are intended to give the UARTn THR FIFO a chance to fill up with data to eliminate many THRE interrupts from occurring at system start-up. The initialization conditions implement a one character delay minus the stop bit whenever THRE = 1 and there have not been at least two characters in the UnTHR at one time since the last THRE = 1 event. This delay is provided to give the CPU time to write data to UnTHR without a THRE interrupt to decode and service. A THRE interrupt is set immediately if the UARTn THR FIFO has held two or more characters at one time and currently, the UnTHR is empty. The THRE interrupt is reset when a UnTHR write occurs or a read of the UnIIR occurs and the THRE is the highest interrupt (UnIIR[3:1] = 001).

14.4.6 UARTn FIFO Control Register (U0FCR - 0x4000 C008, U2FCR - 0x4009 8008, U3FCR - 0x4009 C008)

The write-only UnFCR controls the operation of the UARTn Rx and TX FIFOs.

Table 279: UARTn FIFO Control Register (U0FCR - address 0x4000 C008, U2FCR - 0x4009 8008, U3FCR - 0x4009 C008) bit description

Bit	Symbol	Value	Description	Reset Value
0	FIFO Enable	0	UARTn FIFOs are disabled. Must not be used in the application.	0
		1	Active high enable for both UARTn Rx and TX FIFOs and UnFCR[7:1] access. This bit must be set for proper UART operation. Any transition on this bit will automatically clear the related UART FIFOs.	
1	RX FIFO Reset	0	No impact on either of UARTn FIFOs.	0
		1	Writing a logic 1 to UnFCR[1] will clear all bytes in UARTn Rx FIFO, reset the pointer logic. This bit is self-clearing.	
2	TX FIFO Reset	0	No impact on either of UARTn FIFOs.	0
		1	Writing a logic 1 to UnFCR[2] will clear all bytes in UARTn TX FIFO, reset the pointer logic. This bit is self-clearing.	
3	DMA Mode Select		When the FIFO enable bit (bit 0 of this register) is set, this bit selects the DMA mode. See Section 14.4.6.1.	0
5:4	-	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
7:6	RX Trigger Level		These two bits determine how many receiver UARTn FIFO characters must be written before an interrupt or DMA request is activated.	0
		00	Trigger level 0 (1 character or 0x01)	
		01	Trigger level 1 (4 characters or 0x04)	
		10	Trigger level 2 (8 characters or 0x08)	
		11	Trigger level 3 (14 characters or 0x0E)	
31:8	-		Reserved, user software should not write ones to reserved bits.	NA

14.4.6.1 DMA Operation

The user can optionally operate the UART transmit and/or receive using DMA. The DMA mode is determined by the DMA Mode Select bit in the FCR register. This bit only has an affect when the FIFOs are enabled via the FIFO Enable bit in the FCR register.

UART receiver DMA

In DMA mode, the receiver DMA request is asserted on the event of the receiver FIFO level becoming equal to or greater than trigger level, or if a character time-out occurs. See the description of the RX Trigger Level above. The receiver DMA request is cleared by the DMA controller.

UART transmitter DMA

In DMA mode, the transmitter DMA request is asserted on the event of the transmitter FIFO transitioning to not full. The transmitter DMA request is cleared by the DMA controller.

14.4.7 UARTn Line Control Register (U0LCR - 0x4000 C00C, U2LCR - 0x4009 800C, U3LCR - 0x4009 C00C)

The UnLCR determines the format of the data character that is to be transmitted or received.

Table 280: UARTn Line Control Register (U0LCR - address 0x4000 C00C, U2LCR - 0x4009 800C, U3LCR - 0x4009 C00C) bit description

Bit	Symbol	Value	Description	Reset Value
1:0	Word Length Select	00	5-bit character length	0
		01	6-bit character length	
		10	7-bit character length	
		11	8-bit character length	
2	Stop Bit Select	0	1 stop bit.	0
		1	2 stop bits (1.5 if UnLCR[1:0]=00).	
3	Parity Enable	0	Disable parity generation and checking.	0
		1	Enable parity generation and checking.	
5:4	Parity Select	00	Odd parity. Number of 1s in the transmitted character and the attached parity bit will be odd.	0
		01	Even Parity. Number of 1s in the transmitted character and the attached parity bit will be even.	
		10	Forced "1" stick parity.	
		11	Forced "0" stick parity.	
6	Break Control	0	Disable break transmission.	0
		1	Enable break transmission. Output pin UARTn TXD is forced to logic 0 when UnLCR[6] is active high.	
7	Divisor Latch Access Bit (DLAB)	0	Disable access to Divisor Latches.	0
		1	Enable access to Divisor Latches.	
31:8	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

14.4.8 UARTn Line Status Register (U0LSR - 0x4000 C014, U2LSR - 0x4009 8014, U3LSR - 0x4009 C014)

The UnLSR is a read-only register that provides status information on the UARTn TX and RX blocks.

Table 281: UARTn Line Status Register (U0LSR - address 0x4000 C014, U2LSR - 0x4009 8014, U3LSR - 0x4009 C014) bit description

Bit	Symbol	Value	Description	Reset Value
0	Receiver Data Ready (RDR)		UnLSR0 is set when the UnRBR holds an unread character and is cleared when the UARTn RBR FIFO is empty.	0
		0	The UARTn receiver FIFO is empty.	
		1	The UARTn receiver FIFO is not empty.	
1	Overrun Error (OE)		The overrun error condition is set as soon as it occurs. An UnLSR read clears UnLSR1. UnLSR1 is set when UARTn RSR has a new character assembled and the UARTn RBR FIFO is full. In this case, the UARTn RBR FIFO will not be overwritten and the character in the UARTn RSR will be lost.	0
		0	Overrun error status is inactive.	
		1	Overrun error status is active.	
2	Parity Error (PE)		When the parity bit of a received character is in the wrong state, a parity error occurs. An UnLSR read clears UnLSR[2]. Time of parity error detection is dependent on UnFCR[0]. **Note:** A parity error is associated with the character at the top of the UARTn RBR FIFO.	0
		0	Parity error status is inactive.	
		1	Parity error status is active.	
3	Framing Error (FE)		When the stop bit of a received character is a logic 0, a framing error occurs. An UnLSR read clears UnLSR[3]. The time of the framing error detection is dependent on UnFCR0. Upon detection of a framing error, the Rx will attempt to resynchronize to the data and assume that the bad stop bit is actually an early start bit. However, it cannot be assumed that the next received byte will be correct even if there is no Framing Error. **Note:** A framing error is associated with the character at the top of the UARTn RBR FIFO.	0
		0	Framing error status is inactive.	
		1	Framing error status is active.	
4	Break Interrupt (BI)		When RXDn is held in the spacing state (all zeroes) for one full character transmission (start, data, parity, stop), a break interrupt occurs. Once the break condition has been detected, the receiver goes idle until RXDn goes to marking state (all ones). An UnLSR read clears this status bit. The time of break detection is dependent on UnFCR[0]. **Note:** The break interrupt is associated with the character at the top of the UARTn RBR FIFO.	0
		0	Break interrupt status is inactive.	
		1	Break interrupt status is active.	
5	Transmitter Holding Register Empty (THRE))		THRE is set immediately upon detection of an empty UARTn THR and is cleared on a UnTHR write.	1
		0	UnTHR contains valid data.	
		1	UnTHR is empty.	
6	Transmitter Empty (TEMT)		TEMT is set when both UnTHR and UnTSR are empty; TEMT is cleared when either the UnTSR or the UnTHR contain valid data.	1
		0	UnTHR and/or the UnTSR contains valid data.	
		1	UnTHR and the UnTSR are empty.	

Table 281: UARTn Line Status Register (U0LSR - address 0x4000 C014, U2LSR - 0x4009 8014, U3LSR - 0x4009 C014) bit description *…continued*

Bit	Symbol	Value	Description	Reset Value
7	Error in RX FIFO (RXFE)		UnLSR[7] is set when a character with a Rx error such as framing error, parity error or break interrupt, is loaded into the UnRBR. This bit is cleared when the UnLSR register is read and there are no subsequent errors in the UARTn FIFO.	0
		0	UnRBR contains no UARTn RX errors or UnFCR[0]=0.	
		1	UARTn RBR contains at least one UARTn RX error.	
31:8	-		Reserved, the value read from a reserved bit is not defined.	NA

14.4.9 UARTn Scratch Pad Register (U0SCR - 0x4000 C01C, U2SCR - 0x4009 801C U3SCR - 0x4009 C01C)

The UnSCR has no effect on the UARTn operation. This register can be written and/or read at user’s discretion. There is no provision in the interrupt interface that would indicate to the host that a read or write of the UnSCR has occurred.

Table 282: UARTn Scratch Pad Register (U0SCR - address 0x4000 C01C, U2SCR - 0x4009 801C, U3SCR - 0x4009 C01C) bit description

Bit	Symbol	Description	Reset Value
7:0	Pad	A readable, writable byte.	0x00
31:8	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

14.4.10 UARTn Auto-baud Control Register (U0ACR - 0x4000 C020, U2ACR - 0x4009 8020, U3ACR - 0x4009 C020)

The UARTn Auto-baud Control Register (UnACR) controls the process of measuring the incoming clock/data rate for the baud rate generation and can be read and written at user’s discretion.

Table 283: UARTn Auto-baud Control Register (U0ACR - address 0x4000 C020, U2ACR - 0x4009 8020, U3ACR - 0x4009 C020) bit description

Bit	Symbol	Value	Description	Reset value
0	Start		This bit is automatically cleared after auto-baud completion.	0
		0	Auto-baud stop (auto-baud is not running).	
		1	Auto-baud start (auto-baud is running). Auto-baud run bit. This bit is automatically cleared after auto-baud completion.	
1	Mode		Auto-baud mode select bit.	0
		0	Mode 0.	
		1	Mode 1.	
2	AutoRestart	0	No restart.	0
		1	Restart in case of time-out (counter restarts at next UARTn Rx falling edge)	0
7:3	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

Table 283: UARTn Auto-baud Control Register (U0ACR - address 0x4000 C020, U2ACR - 0x4009 8020, U3ACR - 0x4009 C020) bit description *…continued*

Bit	Symbol	Value	Description	Reset value
8	ABEOIntClr		End of auto-baud interrupt clear bit (write-only accessible). Writing a 1 will clear the corresponding interrupt in the UnIIR. Writing a 0 has no impact.	0
9	ABTOIntClr		Auto-baud time-out interrupt clear bit (write-only accessible). Writing a 1 will clear the corresponding interrupt in the UnIIR. Writing a 0 has no impact.	0
31:10	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

14.4.10.1 Auto-baud

The UARTn auto-baud function can be used to measure the incoming baud-rate based on the “AT” protocol (Hayes command). If enabled the auto-baud feature will measure the bit time of the receive data stream and set the divisor latch registers UnDLM and UnDLL accordingly.

Remark: the fractional rate divider is not connected during auto-baud operations, and therefore should not be used when the auto-baud feature is needed.

Auto-baud is started by setting the UnACR Start bit. Auto-baud can be stopped by clearing the UnACR Start bit. The Start bit will clear once auto-baud has finished and reading the bit will return the status of auto-baud (pending/finished).

Two auto-baud measuring modes are available which can be selected by the UnACR Mode bit. In mode 0 the baud-rate is measured on two subsequent falling edges of the UARTn Rx pin (the falling edge of the start bit and the falling edge of the least significant bit). In mode 1 the baud-rate is measured between the falling edge and the subsequent rising edge of the UARTn Rx pin (the length of the start bit).

The UnACR AutoRestart bit can be used to automatically restart baud-rate measurement if a time-out occurs (the rate measurement counter overflows). If this bit is set the rate measurement will restart at the next falling edge of the UARTn Rx pin.

The auto-baud function can generate two interrupts.

- The UnIIR ABTOInt interrupt will get set if the interrupt is enabled (UnIER ABToIntEn is set and the auto-baud rate measurement counter overflows).
- The UnIIR ABEOInt interrupt will get set if the interrupt is enabled (UnIER ABEOIntEn is set and the auto-baud has completed successfully).

The auto-baud interrupts have to be cleared by setting the corresponding UnACR ABTOIntClr and ABEOIntEn bits.

Typically the fractional baud-rate generator is disabled (DIVADDVAL = 0) during auto-baud. However, if the fractional baud-rate generator is enabled (DIVADDVAL > 0), it is going to impact the measuring of UARTn Rx pin baud-rate, but the value of the UnFDR register is not going to be modified after rate measurement. Also, when auto-baud is used, any write to UnDLM and UnDLL registers should be done before UnACR register write. The minimum and the maximum baud rates supported by UARTn are function of pclk, number of data bits, stop bits and parity bits.

$$ratemin = \frac{2 \times PCLK}{16 \times 2^{15}} \leq UARTn_{baudrate} \leq \frac{PCLK}{16 \times (2 + databits + paritybits + stopbits)} = ratemax \quad (3)$$

14.4.10.2 Auto-baud modes

When the software is expecting an “AT” command, it configures the UARTn with the expected character format and sets the UnACR Start bit. The initial values in the divisor latches UnDLM and UnDLM don‘t care. Because of the “A” or “a” ASCII coding (”A" = 0x41, “a” = 0x61), the UARTn Rx pin sensed start bit and the LSB of the expected character are delimited by two falling edges. When the UnACR Start bit is set, the auto-baud protocol will execute the following phases:

1. On UnACR Start bit setting, the baud rate measurement counter is reset and the UARTn UnRSR is reset. The UnRSR baud rate is switch to the highest rate.
2. A falling edge on UARTn Rx pin triggers the beginning of the start bit. The rate measuring counter will start counting pclk cycles optionally pre-scaled by the fractional baud-rate generator.
3. During the receipt of the start bit, 16 pulses are generated on the RSR baud input with the frequency of the (fractional baud-rate pre-scaled) UARTn input clock, guaranteeing the start bit is stored in the UnRSR.
4. During the receipt of the start bit (and the character LSB for mode = 0) the rate counter will continue incrementing with the pre-scaled UARTn input clock (pclk).
5. If Mode = 0 then the rate counter will stop on next falling edge of the UARTn Rx pin. If Mode = 1 then the rate counter will stop on the next rising edge of the UARTn Rx pin.
6. The rate counter is loaded into UnDLM/UnDLL and the baud-rate will be switched to normal operation. After setting the UnDLM/UnDLL the end of auto-baud interrupt UnIIR ABEOInt will be set, if enabled. The UnRSR will now continue receiving the remaining bits of the “A/a” character.

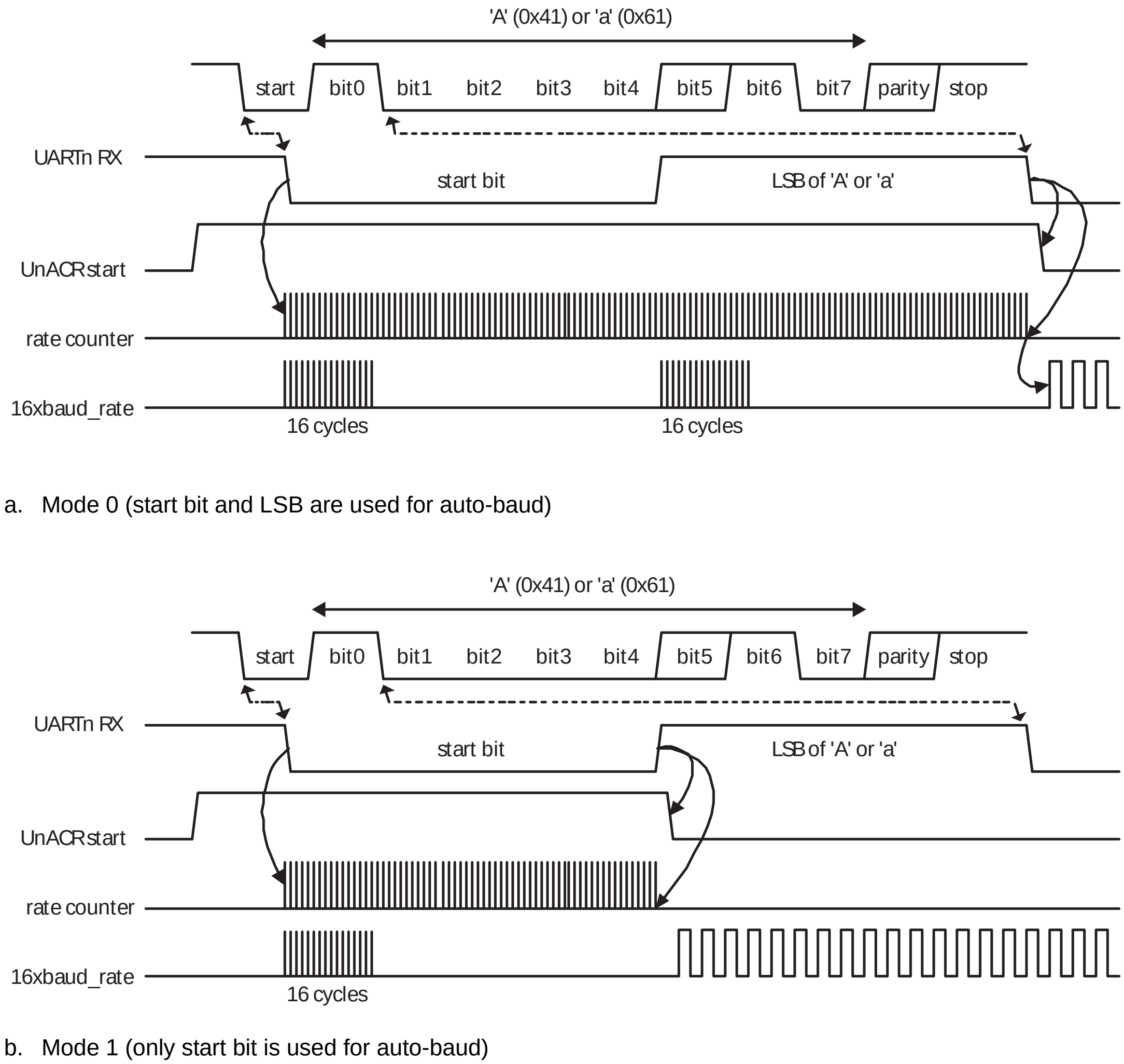

Fig 46. Auto-baud a) mode 0 and b) mode 1 waveform

14.4.11 UARTn IrDA Control Register (U0ICR - 0x4000 C024, U2ICR - 0x4009 8024, U3ICR - 0x4009 C024)

The IrDA Control Register enables and configures the IrDA mode on each UART. The value of UnICR should not be changed while transmitting or receiving data, or data loss or corruption may occur.

Table 284: UARTn IrDA Control Register (U0ICR - 0x4000 C024, U2ICR - 0x4009 8024, U3ICR - 0x4009 C024) bit description

Bit	Symbol	Value	Description	Reset value
0	IrDAEn	0	IrDA mode on UARTn is disabled, UARTn acts as a standard UART.	0
		1	IrDA mode on UARTn is enabled.	
1	IrDAInv		When 1, the serial input is inverted. This has no effect on the serial output. When 0, the serial input is not inverted.	0

Table 284: UARTn IrDA Control Register (U0ICR - 0x4000 C024, U2ICR - 0x4009 8024, U3ICR - 0x4009 C024) bit description *…continued*

Bit	Symbol	Value	Description	Reset value
2	FixPulseEn		When 1, enabled IrDA fixed pulse width mode.	0
5:3	PulseDiv		Configures the pulse when FixPulseEn = 1. See text below for details.	0
31:6	-	NA	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	0

The PulseDiv bits in UnICR are used to select the pulse width when the fixed pulse width mode is used in IrDA mode (IrDAEn = 1 and FixPulseEn = 1). The value of these bits should be set so that the resulting pulse width is at least 1.63 µs. Table 285 shows the possible pulse widths.

Table 285: IrDA Pulse Width

FixPulseEn	PulseDiv	IrDA Transmitter Pulse width (µs)
0	x	3 / (16 × baud rate)
1	0	$2 \times T_{PCLK}$
1	1	$4 \times T_{PCLK}$
1	2	$8 \times T_{PCLK}$
1	3	$16 \times T_{PCLK}$
1	4	$32 \times T_{PCLK}$
1	5	$64 \times T_{PCLK}$
1	6	$128 \times T_{PCLK}$
1	7	$256 \times T_{PCLK}$

14.4.12 UARTn Fractional Divider Register (U0FDR - 0x4000 C028, U2FDR - 0x4009 8028, U3FDR - 0x4009 C028)

The UART0/2/3 Fractional Divider Register (U0/2/3FDR) controls the clock pre-scaler for the baud rate generation and can be read and written at the user’s discretion. This pre-scaler takes the APB clock and generates an output clock according to the specified fractional requirements.

Important: If the fractional divider is active (DIVADDVAL > 0) and DLM = 0, the value of the DLL register must be greater than 2.

Table 286: UARTn Fractional Divider Register (U0FDR - address 0x4000 C028, U2FDR - 0x4009 8028, U3FDR - 0x4009 C028) bit description

Bit	Function	Value	Description	Reset value
3:0	DIVADDVAL	0	Baud-rate generation pre-scaler divisor value. If this field is 0, fractional baud-rate generator will not impact the UARTn baudrate.	0
7:4	MULVAL	1	Baud-rate pre-scaler multiplier value. This field must be greater or equal 1 for UARTn to operate properly, regardless of whether the fractional baud-rate generator is used or not.	1
31:8	-	NA	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	0

This register controls the clock pre-scaler for the baud rate generation. The reset value of the register keeps the fractional capabilities of UART0/2/3 disabled making sure that UART0/2/3 is fully software and hardware compatible with UARTs not equipped with this feature.

UART0/2/3 baud rate can be calculated as (n = 0/2/3):

$$UARTn_{baudrate} = \frac{PCLK}{16 \times (256 \times UnDLM + UnDLL) \times \left(1 + \frac{DivAddVal}{MulVal}\right)} \quad (4)$$

Where PCLK is the peripheral clock, U0/2/3DLM and U0/2/3DLL are the standard UART0/2/3 baud rate divider registers, and DIVADDVAL and MULVAL are UART0/2/3 fractional baud rate generator specific parameters.

The value of MULVAL and DIVADDVAL should comply to the following conditions:

1. 1 ≤ MULVAL ≤ 15
2. 0 ≤ DIVADDVAL ≤ 14
3. DIVADDVAL < MULVAL

The value of the U0/2/3FDR should not be modified while transmitting/receiving data or data may be lost or corrupted.

If the U0/2/3FDR register value does not comply to these two requests, then the fractional divider output is undefined. If DIVADDVAL is zero then the fractional divider is disabled, and the clock will not be divided.

14.4.12.1 Baud rate calculation

UARTn can operate with or without using the Fractional Divider. In real-life applications it is likely that the desired baud rate can be achieved using several different Fractional Divider settings. The following algorithm illustrates one way of finding a set of DLM, DLL, MULVAL, and DIVADDVAL values. Such set of parameters yields a baud rate with a relative error of less than 1.1% from the desired one.

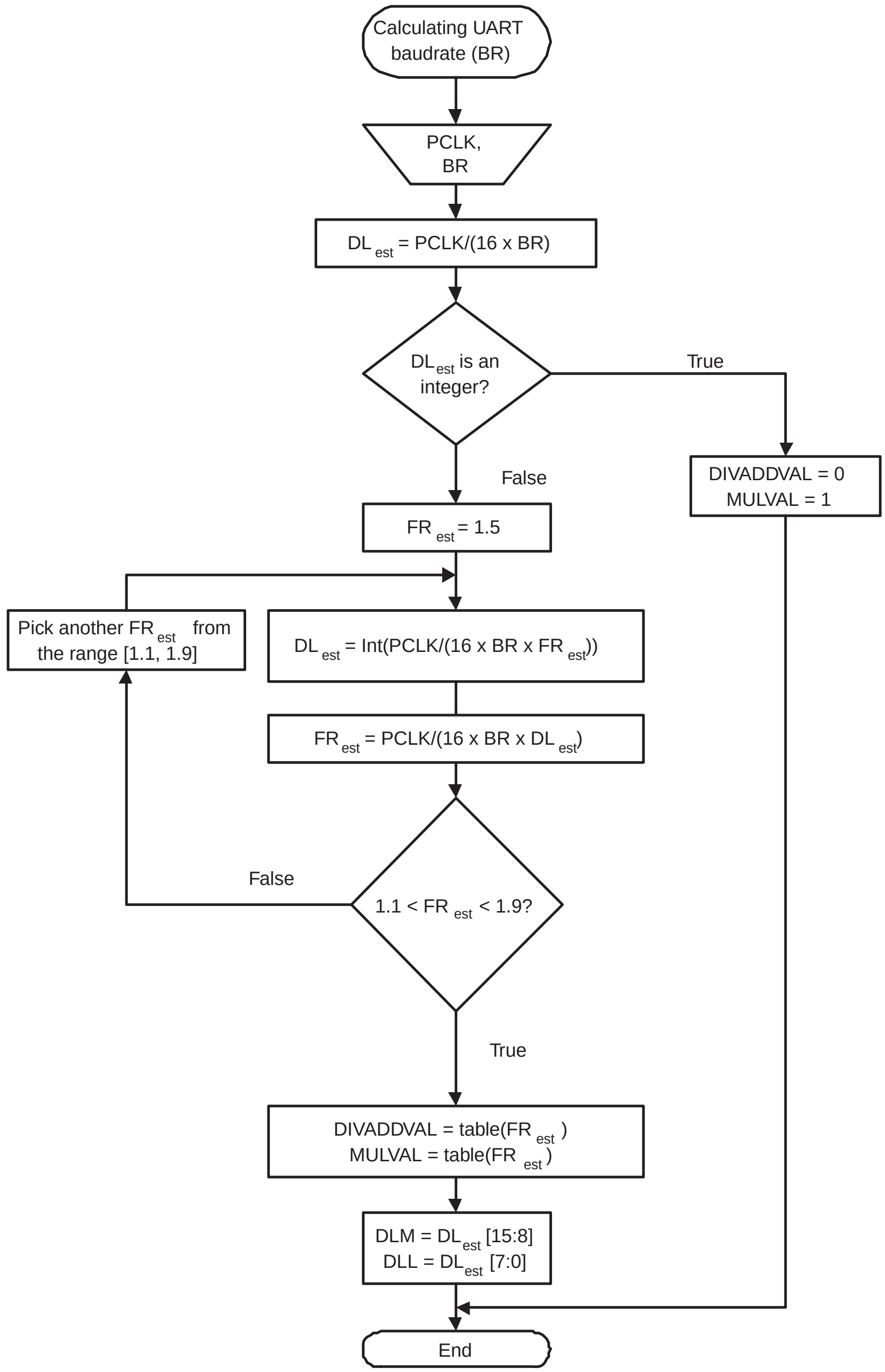

Fig 47. Algorithm for setting UART dividers

Table 287. Fractional Divider setting look-up table

FR	DivAddVal/ MulVal	FR	DivAddVal/ MulVal	FR	DivAddVal/ MulVal	FR	DivAddVal/ MulVal
1.000	0/1	1.250	1/4	1.500	1/2	1.750	3/4
1.067	1/15	1.267	4/15	1.533	8/15	1.769	10/13
1.071	1/14	1.273	3/11	1.538	7/13	1.778	7/9
1.077	1/13	1.286	2/7	1.545	6/11	1.786	11/14
1.083	1/12	1.300	3/10	1.556	5/9	1.800	4/5
1.091	1/11	1.308	4/13	1.571	4/7	1.818	9/11
1.100	1/10	1.333	1/3	1.583	7/12	1.833	5/6
1.111	1/9	1.357	5/14	1.600	3/5	1.846	11/13
1.125	1/8	1.364	4/11	1.615	8/13	1.857	6/7
1.133	2/15	1.375	3/8	1.625	5/8	1.867	13/15
1.143	1/7	1.385	5/13	1.636	7/11	1.875	7/8
1.154	2/13	1.400	2/5	1.643	9/14	1.889	8/9
1.167	1/6	1.417	5/12	1.667	2/3	1.900	9/10
1.182	2/11	1.429	3/7	1.692	9/13	1.909	10/11
1.200	1/5	1.444	4/9	1.700	7/10	1.917	11/12
1.214	3/14	1.455	5/11	1.714	5/7	1.923	12/13
1.222	2/9	1.462	6/13	1.727	8/11	1.929	13/14
1.231	3/13	1.467	7/15	1.733	11/15	1.933	14/15

14.4.12.1.1 Example 1: PCLK = 14.7456 MHz, BR = 9600

According to the provided algorithm DL_{est} = PCLK/(16 x BR) = 14.7456 MHz / (16 x 9600) = 96. Since this DL_{est} is an integer number, DIVADDVAL = 0, MULVAL = 1, DLM = 0, and DLL = 96.

14.4.12.1.2 Example 2: PCLK = 12 MHz, BR = 115200

According to the provided algorithm DL_{est} = PCLK/(16 x BR) = 12 MHz / (16 x 115200) = 6.51. This DL_{est} is not an integer number and the next step is to estimate the FR parameter. Using an initial estimate of FR_{est} = 1.5 a new DL_{est} = 4 is calculated and FR_{est} is recalculated as FR_{est} = 1.628. Since FRest = 1.628 is within the specified range of 1.1 and 1.9, DIVADDVAL and MULVAL values can be obtained from the attached look-up table.

The closest value for FRest = 1.628 in the look-up Table 287 is FR = 1.625. It is equivalent to DIVADDVAL = 5 and MULVAL = 8.

Based on these findings, the suggested UART setup would be: DLM = 0, DLL = 4, DIVADDVAL = 5, and MULVAL = 8. According to Equation 4 the UART rate is 115384. This rate has a relative error of 0.16% from the originally specified 115200.

14.4.13 UARTn Transmit Enable Register (U0TER - 0x4000 C030, U2TER - 0x4009 8030, U3TER - 0x4009 C030)

The UnTER register enables implementation of software flow control. When TXEn=1, UARTn transmitter will keep sending data as long as they are available. As soon as TXEn becomes 0, UARTn transmission will stop.

Table 288 describes how to use TXEn bit in order to achieve software flow control.

Table 288: UARTn Transmit Enable Register (U0TER - address 0x4000 C030, U2TER - 0x4009 8030, U3TER - 0x4009 C030) bit description

Bit	Symbol	Description	Reset Value
6:0	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
7	TXEN	When this bit is 1, as it is after a Reset, data written to the THR is output on the TXD pin as soon as any preceding data has been sent. If this bit is cleared to 0 while a character is being sent, the transmission of that character is completed, but no further characters are sent until this bit is set again. In other words, a 0 in this bit blocks the transfer of characters from the THR or TX FIFO into the transmit shift register. Software implementing software-handshaking can clear this bit when it receives an XOFF character (DC3). Software can set this bit again when it receives an XON (DC1) character.	1

14.5 Architecture

The architecture of the UARTs 0, 2 and 3 are shown below in the block diagram.

The APB interface provides a communications link between the CPU or host and the UART.

The UARTn receiver block, UnRX, monitors the serial input line, RXDn, for valid input. The UARTn RX Shift Register (UnRSR) accepts valid characters via RXDn. After a valid character is assembled in the UnRSR, it is passed to the UARTn RX Buffer Register FIFO to await access by the CPU or host via the generic host interface.

The UARTn transmitter block, UnTX, accepts data written by the CPU or host and buffers the data in the UARTn TX Holding Register FIFO (UnTHR). The UARTn TX Shift Register (UnTSR) reads the data stored in the UnTHR and assembles the data to transmit via the serial output pin, TXDn.

The UARTn Baud Rate Generator block, UnBRG, generates the timing enables used by the UARTn TX block. The UnBRG clock input source is the APB clock (PCLK). The main clock is divided down per the divisor specified in the UnDLL and UnDLM registers. This divided down clock is the 16x oversample clock.

The interrupt interface contains registers UnIER and UnIIR. The interrupt interface receives several one clock wide enables from the UnTX and UnRX blocks.

Status information from the UnTX and UnRX is stored in the UnLSR. Control information for the UnTX and UnRX is stored in the UnLCR.

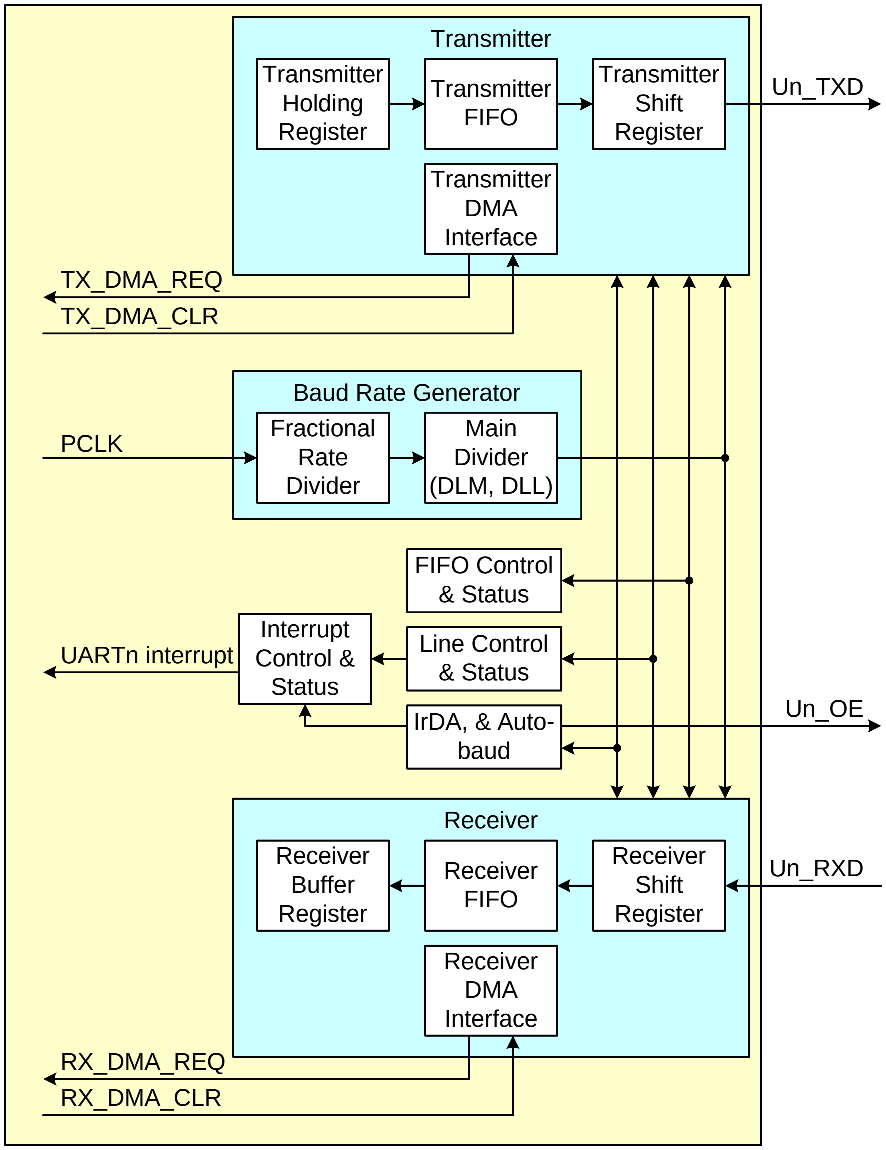

Fig 48. UART0, 2 and 3 block diagram